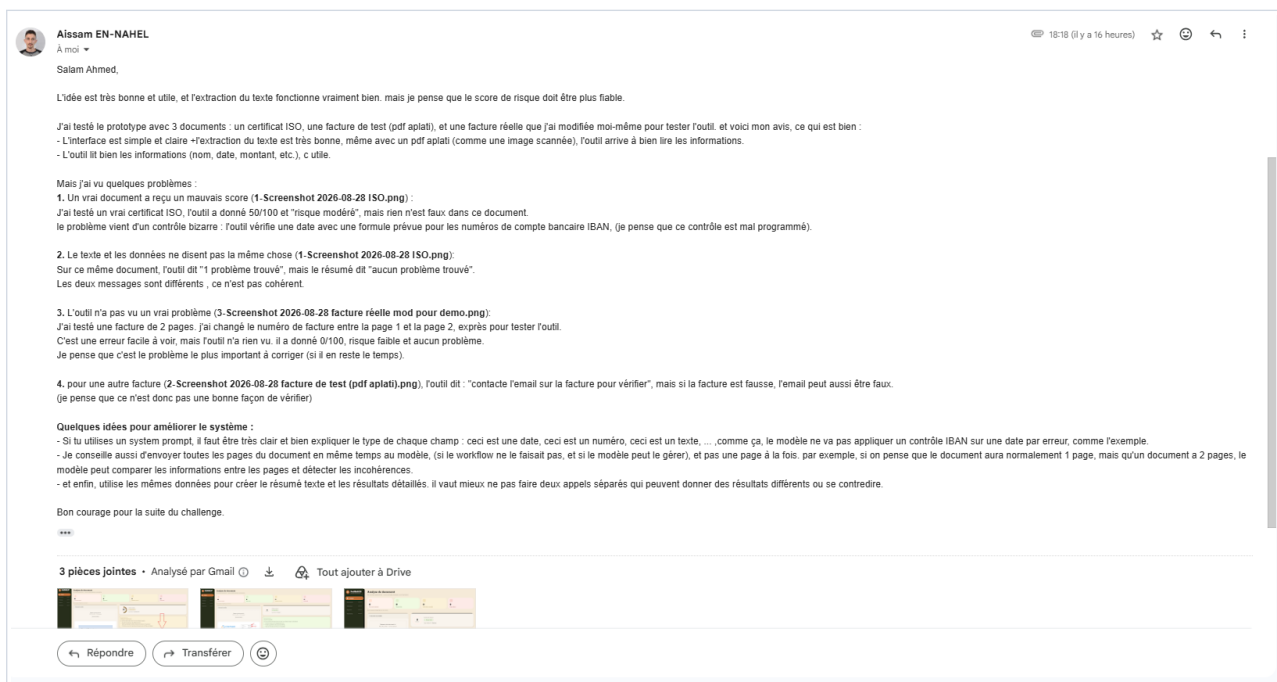


## 1. First external validation — TecForge

|          |                                                                                                                                                                    |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company  | TecForge                                                                                                                                                           |
| Tester   | Aissam EN-NAHEL, Human Resources — written feedback, 28 August 2026                                                                                                |
| Material | Three documents: a genuine ISO certificate, a flattened test invoice, and a real two-page invoice the tester altered himself to see whether the tool would notice. |
| Method   | Unassisted use of the deployed prototype, followed by written feedback. No script, no guidance.                                                                    |

### Verdict in the tester's own words

The idea was judged useful and the text extraction very good — including on a flattened PDF read as a scanned image. *“Je pense que le score de risque doit être plus fiable”*: the concern was the reliability of the risk score, not the reading of the document.



Feedback received from Aissam EN-NAHEL (TecForge), reproduced verbatim.

### Held up under test

- ✓ Interface judged simple and clear.
- ✓ Extraction accurate even on a flattened PDF.
- ✓ Fields read correctly: name, date, amount.

### Did not hold up

- ✗ A genuine document scored as suspicious.
- ✗ Narrative contradicted the anomaly count.
- ✗ A deliberate forgery went unnoticed.

## 2. The four defects, and their cause in the code

Each report was traced to a specific line before anything was changed. None was a tuning problem; all four were logic errors.

- 1. A genuine ISO certificate scored 50/100.** The IBAN checksum fired on the shape of a value while ignoring its label. “Two letters, two digits, then 11 to 30 characters” describes a certificate number as well as a bank account, so the mod-97 key failed on an authentic document — and a single failed check imposed a score floor of 50, turning a false positive into “moderate risk”.
- 2. The count said one problem, the summary said none.** The model writes its explanation before the deterministic checks run. It could therefore report “nothing found” while those checks were adding an anomaly to the list. Two true statements about two different stages, printed as one contradiction.
- 3. An invoice number changed between page 1 and page 2 went unseen.** The PDF renderer read `getPage(1)` only, while reporting the real page count. The altered page never reached the model. The tester ranked this the most important defect; the diagnosis confirms it, since no amount of prompt tuning could have recovered a page

that was never sent.

4. **“Contact the email on the invoice to verify.”** A forged document carries forged contact details. The prompt asked for “concrete verification steps” without ever forbidding the document from vouching for itself.

3. From feedback to shipped change

Every point raised was corrected and deployed within the challenge window, not deferred to a backlog. The corresponding changes are public in the project repository: [github.com/Hajji-ahmed/TrustDoc-AI](https://github.com/Hajji-ahmed/TrustDoc-AI).

| Reported                                        | Change made                                                                                                                                                                                               | Status    |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Wrong rule applied to a date                    | The IBAN check now requires a label that names a bank account. An IBAN filed under a label that does not say so escapes the check — accusing a genuine document costs more than missing one verification. | SHIPPED   |
| Summary contradicts detail                      | Each statement is attributed to its stage: visual analysis first, automatic checks second. The model’s prose is kept, not rewritten.                                                                      | SHIPPED   |
| Two pages never compared                        | All pages are rendered and sent to the model in one request, each announced by its number. The prompt requires values repeated across pages to be compared.                                               | SHIPPED   |
| Email alone is weak evidence                    | The prompt forbids any verification routed through the document’s own contact details, and requires an independent source instead.                                                                        | SHIPPED   |
| Field types left implicit (tester’s suggestion) | The prompt now states which labels may be used for which value types, so the wrong deterministic check cannot be triggered downstream.                                                                    | SHIPPED   |
| Cross-page detection on the tester’s own file   | Route contract verified on six cases; the two-page invoice has not been run again through the model.                                                                                                      | TO RETEST |

What this round actually taught

Extraction quality was never the weak point — the tester confirmed it works, including on flattened scans. Every defect he found sat in the layer that decides *what the reading means*: which rule applies to which field, whether pages are compared, whether the narrative matches the findings. A screening tool is not limited by what it can read, but by the discipline of what it concludes.

The correction that has not yet been proven

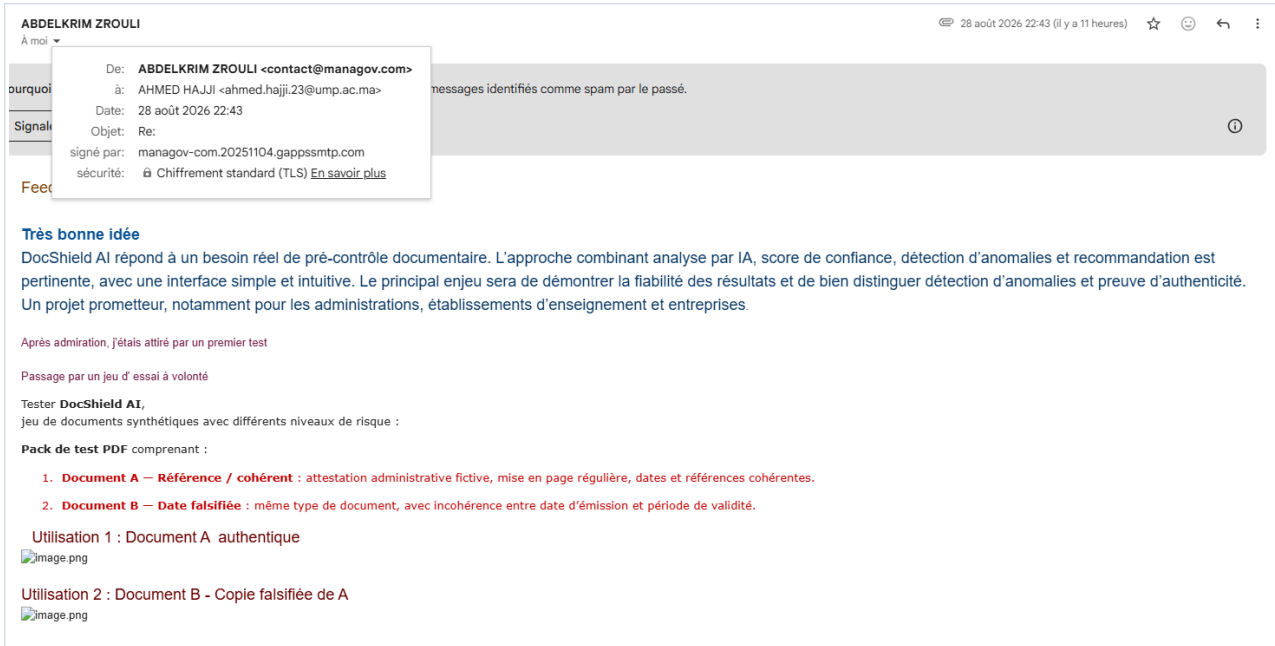
Points 1, 2 and 4 are verified by automated assertions. Point 3 — the one the tester ranked highest — is verified only up to the request contract: all pages now reach the model. Whether the model *notices* the altered invoice number requires re-running his own two-page file. Until that is done, this defect is fixed in the pipeline and unproven in the result.

#### 4. Second external validation — Managov

|          |                                                                                                                                                                                              |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company  | Managov                                                                                                                                                                                      |
| Reviewer | Abdelkrim Zrouli, Director — written feedback, 28 August 2026                                                                                                                                |
| Material | A synthetic test pack he built himself: Document A, a coherent administrative attestation, and Document B, the same document with its issue date made inconsistent with its validity period. |
| Method   | Unprompted assessment of the concept, followed by a controlled A/B test on a matched pair of documents.                                                                                      |

##### Assessment

*“DocShield AI répond à un besoin réel de pré-contrôle documentaire.”* The combination of AI analysis, confidence score, anomaly detection and recommendation was judged relevant, the interface simple and intuitive, and the project promising for public administrations, educational institutions and companies.



Feedback received from Abdelkrim Zrouli (Managov), reproduced verbatim.

##### The reservation, and why it matters more than the praise

*“Le principal enjeu sera de démontrer la fiabilité des résultats et de bien distinguer détection d’anomalies et preuve d’authenticité.”*

This is the product’s own founding constraint, stated back by someone who had not been told it. DocShield never claims a document is authentic; it reports what looks irregular and hands the decision to a human. Hearing that boundary named as *the* central challenge, by a director evaluating the tool for his own organisation, confirms that the distinction is not academic — it is what buyers will judge the product on.

#### 4.1 The test design — one controlled variable

Both documents are byte-for-byte identical except for the validity window. That is a cleaner experiment than anything run in-house: it isolates a single failure mode and ships with its own control.

| Field           | Document A — control                                                         | Document B — altered             |
|-----------------|------------------------------------------------------------------------------|----------------------------------|
| Issue date      | 28 August 2026                                                               | 28 August 2026                   |
| Period covered  | 1 Jan 2026 to 31 Dec 2026                                                    | 1 Jan 2026 to <b>31 Dec 2025</b> |
| Validity        | until 31 Dec 2026                                                            | until <b>31 Dec 2025</b>         |
| Everything else | Identical — reference, identifier, issuer, signature, verification reference |                                  |

The planted defect is a period that ends before it begins, on a document issued after both dates. It is obvious to a human and invisible to any check that reads fields one at a time.

## 4.2 The result — the planted defect was caught

### Document B, scored by the deployed prototype

**Score** 34/100 — moderate risk. **Type** Attestation administrative, correctly identified.

**Anomaly** One, and only one: “*La période concernée commence en 2026 mais termine en 2025, ce qui est chronologiquement impossible.*” Classified under data coherence, moderate severity, and drawn as a numbered region on the document itself.

**Recommendation** Manual verification required — confirm the details with the issuer *through alternative official channels*, and compare against other documents.

### What this result confirms beyond the detection itself

Two of the corrections made after the first validation are visible in this output, independently produced by a second reviewer who knew nothing of them.

**The narrative no longer contradicts the findings.** The explanation reads “*Analyse visuelle: . . .*” then “*Contrôles automatiques: aucun écart relevé. . .*”. Each statement is attributed to its stage, and both are true at once: the model saw the date problem, the deterministic checks correctly found nothing in their own scope.

**The verification step no longer routes through the document.** The recommendation names *alternative official channels* rather than the contact details printed on the page — the precise flaw reported by the first reviewer.

### The control was not scored

Document A exists to answer the question the reviewer wrote at the top of both files: “*Vérifier que DocShield AI ne génère pas de faux positifs.*” It was uploaded, but the captured screen shows it before the analysis was run, so **no result is recorded for the control**.

Detection on B is therefore established; the absence of a false positive on A is not. A test that only ever fires proves nothing on its own — the control is what separates a detector from an alarm that is always on. Running Document A is the single most useful next step in this dossier.

## 5. Validation register

One line per company contacted during the challenge. Only verified conversations and real feedback are recorded here; the blank rows are reserved for the validations still in progress.

| Company  | Person / Role                             | Feedback / Learning                                                                                                                                                                                                                                                                                                                                               | Change / Next action                                                                                                                                                          |
|----------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TecForge | Aissam EN-NAHEL<br><i>Human Resources</i> | Document reading judged reliable, including on flattened scans; the risk assessment was not. Four defects reported: a validation rule applied to the wrong type of field, a summary inconsistent with the detailed findings, no comparison between the pages of a single document, and a verification step relying on the document's own contact details.         | All four defects corrected and deployed within the challenge window. Next: retest the altered two-page invoice, then extend testing to other document families.               |
| Managov  | Abdelkrim Zrouli<br><i>Director</i>       | Concept and interface judged relevant for administrations, schools and companies. Central challenge named: demonstrating the reliability of results, and keeping anomaly detection clearly distinct from proof of authenticity. Ran a matched pair of synthetic documents differing by one falsified validity window; the altered document was correctly flagged. | Positioning confirmed and kept explicit in the interface. Detection validated on the altered document. Next: run the control document to confirm no false positive is raised. |
|          |                                           |                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                               |
|          |                                           |                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                               |

## 6. Summary and evidence checklist

**Completed** Two external companies, two reviewers — one HR practitioner and one director — seven documents tested, four defects found and four corrections shipped within the challenge window.

**Confirmed from outside** The boundary the product was built on — report anomalies, never certify authenticity — was independently named as the central challenge by the second reviewer, who had not been told it. Two of the corrections made after the first validation are visible in his test output.

**Detection demonstrated** On a falsification designed by a third party, the prototype isolated the planted defect, named it correctly and localised it on the document.

**Not yet demonstrated** The absence of false positives. The control document of that same test was never scored, and the two-page invoice from the first validation has not been re-run.

**Next step** Run Document A, the control. It is the cheapest test remaining and the only one that turns the current evidence into a complete result.

## Evidence checklist

- ✓ Two companies contacted, both replied with written feedback
- ✓ Both messages captured and reproduced verbatim
- ✓ Root cause identified in code for all four defects
- ✓ Corrections implemented and deployed, published in the project repository
- ✓ Detection confirmed on a third party's falsified document
- ✗ Control document not scored — absence of false positives unverified
- ✗ Retest of the two-page invoice with the corrected prototype